English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

English transcription of the handwritten solution notes. The calculations and original level of detail are retained; apparent source slips are identified in the transcription notes.

Problem 1

1. Conditional probability

There are two yellow, three green and four blue balls. A yellow ball means rolling one die, green means two dice, and blue means three dice. By Bayes' formula,

$$\begin{aligned}\mathbb{P}(\text{green} \mid \text{sum } 3) &= \frac{\mathbb{P}(\text{sum } 3 \mid \text{green})\mathbb{P}(\text{green})}{\mathbb{P}(\text{sum } 3)} \\ &= \frac{\frac{2}{36}\cdot\frac{3}{9}}{\frac{1}{6}\cdot\frac{2}{9}+\frac{2}{36}\cdot\frac{3}{9}+\frac{1}{216}\cdot\frac{4}{9}} = \frac{9}{28} \approx 32.14\%.\end{aligned}$$

2. Five dice

$$\mathbb{P}(\max \geq 3) = 1 - \mathbb{P}(\text{all outcomes are 1 or 2}) = 1 - \left(\frac{2}{6}\right)^5.$$

Let C be the event that at least one number is divisible by 5, and D the event that at least one number is divisible by 2. Then

$$\mathbb{P}(C^c) = \left(\frac{5}{6}\right)^5, \qquad \mathbb{P}(D^c) = \left(\frac{1}{2}\right)^5, \qquad \mathbb{P}(C^c \cap D^c) = \left(\frac{2}{6}\right)^5.$$

We seek

$$\begin{aligned}\mathbb{P}(C \cap D) &= \mathbb{P}(C) + \mathbb{P}(D) - \mathbb{P}(C \cup D) \\ &= 1 - \mathbb{P}(C^c) + 1 - \mathbb{P}(D^c) - (1 - \mathbb{P}(C^c \cap D^c)) \\ &= 1 - \left(\frac{5}{6}\right)^5 - \left(\frac{1}{2}\right)^5 + \left(\frac{1}{6}\right)^5 \approx 56.70\%. \qquad \text{(Note 1)}\end{aligned}$$

3. Seven balls in seven boxes

Two boxes contain three balls each and one box contains one ball when $X_3 = 2$, so

$$\mathbb{P}(X_3 = 2) = \frac{7\binom{7}{3}\,6\binom{4}{3}\,5/2}{7^7} \approx 1.78\%.$$

When $X_3 = 1$, the nonzero occupancies are $(3,4)$, $(3,1,1,1,1)$, $(3,2,2)$ or $(3,2,1,1)$. Thus

$$\begin{aligned}\mathbb{P}(X_3 = 1) = &\frac{7\binom{7}{3}\,6}{7^7} + \frac{7\binom{7}{3}\,6\cdot5\cdot4\cdot3}{7^7} \\ &+ \frac{7\binom{7}{3}\,6\binom{4}{2}\,5/2}{7^7} + \frac{7\binom{7}{3}\,6\binom{4}{2}\,5\cdot4}{7^7} \approx 34.99\%.\end{aligned}$$

The source then writes

$$\mathbb{P}(X_3 = 0) = 1 - \mathbb{P}(X_3 = 0 \text{ or } 1) \approx 63.23\%. \qquad \text{(Note 2)}$$

Consequently,

$$\mathbb{E}X_3 = \mathbb{P}(X_3 = 1) + 2\mathbb{P}(X_3 = 2) \approx 0.3855.$$

Also,

$$\sum_{i=1}^{7} i\mathbb{E}X_i = \mathbb{E}\left[\sum_{i=0}^{7} iX_i\right] = \mathbb{E}[\text{number of balls}] = 7.$$

Similarly,

$$\sum_{i=0}^{7} \mathbb{E}X_i = \mathbb{E}[X_0 + \cdots + X_7] = \mathbb{E}[\text{number of boxes}] = 7.$$

Problem 2

Let Y be the number of correct answers among the questions the student knows, and Z the number of correct answers among those the student does not know. Then

$$Y \sim \mathrm{Bin}(10, 9/10), \qquad Z \sim \mathrm{Bin}(5, 1/5), \qquad X = Y + Z.$$

1. More than twelve correct answers; moments

$$\begin{aligned}\mathbb{P}(X > 12) &= \sum_{k=3}^{5} \mathbb{P}(Z = k)\mathbb{P}(Y > 12 - k) \\ &= \mathbb{P}(Z = 3)\mathbb{P}(Y = 10) \\ &\quad + \mathbb{P}(Z = 4)[\mathbb{P}(Y = 10) + \mathbb{P}(Y = 9)] \\ &\quad + \mathbb{P}(Z = 5)[\mathbb{P}(Y = 10) + \mathbb{P}(Y = 9) + \mathbb{P}(Y = 8)].\end{aligned}$$

$$\mathbb{E}X = \mathbb{E}Y + \mathbb{E}Z = 10 \cdot \frac{9}{10} + 5 \cdot \frac{1}{5} = 10.$$

By independence,

$$DX = DY + DZ = 10 \cdot \frac{9}{10} \cdot \frac{1}{10} + 5 \cdot \frac{1}{5} \cdot \frac{4}{5} = \frac{17}{10}.$$

2. Conditional probability

By Bayes' formula,

$$\begin{aligned}\mathbb{P}(Y = 3 \mid X = 3) &= \frac{\mathbb{P}(X = 3 \mid Y = 3)\mathbb{P}(Y = 3)}{\sum_k \mathbb{P}(X = 3 \mid Y = k)\mathbb{P}(Y = k)} \\ &= \frac{\mathbb{P}(Z = 0)\mathbb{P}(Y = 3)}{\sum_{k=0}^{3} \mathbb{P}(Z = 3 - k)\mathbb{P}(Y = k)} \\ &= \frac{(4/5)^5 \binom{10}{3}(9/10)^3(1/10)^7}{\displaystyle\sum_{k=0}^{3} \binom{5}{3-k}(1/5)^{3-k}(4/5)^{k+2}\binom{10}{k}(9/10)^k(1/10)^{10-k}} = \cdots.\end{aligned}$$

Problem 3

1. Joint distribution

$Z \backslash X$	0	1	2	$\mathbb{P}(Z = z)$
0	1/9	0	2/9	1/3
1	2/9	1/9	0	1/3
2	0	2/9	1/9	1/3
$\mathbb{P}(X = x)$	1/3	1/3	1/3	1

The source calls the second variable Y in the following calculation (the variable denoted Z in the table):

$$\mathbb{E}X = \mathbb{E}Y = 1, \qquad \mathbb{E}[XY] = 1 \cdot 1 \cdot \frac{1}{9} + 2 \cdot 1 \cdot \frac{2}{9} + 2 \cdot 2 \cdot \frac{1}{9} = 1.$$

Hence $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}X\mathbb{E}Y = 0$ and $\mathrm{Cor}(X, Y) = 0$.

2. Generating function

$$\begin{aligned}g_X(s) &= \tfrac{1}{6}(s + \cdots + s^6), \\ g'_X(s) &= \tfrac{1}{6}(1 + 2s + \cdots + 6s^5), && g'_X(1) = \tfrac{7}{2}, \\ g''_X(s) &= \tfrac{1}{6}(2 + 6s + 12s^2 + 20s^3 + 30s^4), && g''_X(1) = \tfrac{70}{6}.\end{aligned}$$

Thus

$$\mathbb{E}X = g'_X(1) = \frac{7}{2}, \qquad DX = g''_X(1) + g'_X(1) - [g'_X(1)]^2 = \frac{70}{6} + \frac{7}{2} - \frac{49}{4} = \frac{35}{12}.$$

3. Independence

If X, Y are independent, then $\mathbb{P}(X = k, Y = m) = \mathbb{P}(X = k)\mathbb{P}(Y = m)$, so

$$\begin{aligned} g_{X+Y}(s) &= \sum_{k=0}^{\infty} s^k \mathbb{P}(X + Y = k) \\ &= \sum_{k=0}^{\infty} s^k \sum_{m=0}^{k} \mathbb{P}(X = m)\mathbb{P}(Y = k - m) \\ &= \sum_{m=0}^{\infty} \sum_{k=m}^{\infty} s^m \mathbb{P}(X = m) s^{k-m} \mathbb{P}(Y = k - m) \\ &= \sum_{m=0}^{\infty} s^m \mathbb{P}(X = m) \sum_{i=0}^{\infty} s^i \mathbb{P}(Y = i) = g_X(s) g_Y(s), \end{aligned}$$

where $i = k - m$. The converse is false: $g_{X+Y} = g_X g_Y$ does not imply independence. The variables X, Z from part 1 give a counterexample. Verify this!

Problem 4

Let $p_n(k) = \mathbb{P}(k \in E_n)$, the probability that k is selected when the initial choice is from $\{1, \dots, n\}$. The key observation is

$$p_n(k) = \underbrace{p_n(k+1)}_{\substack{\text{select } k \text{ without selecting } k+1 \\ \text{(why is this probability } p_n(k+1)\text{?)}}} + \underbrace{p_n(k+1)\frac{1}{k}}_{\text{select } k+1 \text{ and then } k}.$$

Thus $p_n(k) = \frac{k+1}{k} p_n(k+1)$. Since $p_n(n) = 1/n$,

$$p_n(k) = \frac{k+1}{k}\frac{k+2}{k+1} \cdots \frac{n}{n-1} p_n(n) = \frac{1}{k}.$$

2. Using the preceding observation,

$$\mathbb{P}(2 \in E_n \mid 3 \notin E_n) = \frac{p_n(3)}{1 - p_n(3)} = \frac{1/3}{2/3} = \frac{1}{2}.$$

3. Let $X_i = \mathbf{1}_{\{i \text{ was selected}\}} \sim \text{Ber}(1/i)$. By linearity,

$$\mathbb{E}|E_n| = \mathbb{E}[X_1 + \cdots + X_n] = \sum_{i=1}^{n} \mathbb{E}X_i = 1 + \frac{1}{2} + \cdots + \frac{1}{n}.$$

Remark 1. In fact, X_i and X_j are independent for $i \neq j$!

Remark 2. This procedure is actually the reverse of the Reservoir Sampler (see the exercise classes), which also explains Remark 1.

Transcription notes

1. The source's final term $(1/6)^5$ conflicts with the preceding $(2/6)^5$. The reported 56.70% agrees with $(2/6)^5$.
2. The source writes "$X_3 = 0$ or 1" in the complement. It should refer to $X_3 = 1$ or $X_3 = 2$.